# Girls Build Bots — Resource Guide

For troop leaders and robotics teams running badge workshops

## Official Girl Scout Resources (start here)

- **Volunteer Toolkit (VTK)** — my.girlscouts.org → Volunteer Toolkit. The official, free source for all badge step requirements, meeting agendas, and materials lists. **Always verify workshop activities against the VTK before awarding badges.**
- **Girl Scout Shop** — girlscoutshop.com. Purchase physical badges and the Robotics badge requirement pamphlets for each level.
- **Girl Scout Activity Zone** — girlscouts.org/en/activities-for-girls. Free adapted single-step activities you can preview publicly.
- **Girl Scouts of Greater Atlanta** — girlscoutsatl.org. Local council programming, STEM events, and Gold Award resources.

## Free Programming Platforms (no accounts required to start)

- **MakeCode Arcade** — arcade.makecode.com. Block-based game programming in the browser; the primary platform in this curriculum. Works on any laptop, tablet, or Chromebook. Also has a full curriculum section for educators.
- **Scratch** — scratch.mit.edu. MIT's free block-based programming community; great follow-on for Juniors.
- **ScratchJr** — scratchjr.org. Tablet app for younger Brownies who aren't reading fluently yet.
- **Code.org** — code.org. Free structured courses and Hour of Code activities for all ages; teacher dashboards available.
- **micro:bit** — microbit.org. Browser simulator is free; physical boards (~$20) bridge from screen coding to real hardware.

## Hardware & Robot Kits (optional — by budget)

| Budget | Options |
|---|---|
| $0 | MakeCode Arcade / Scratch simulators, unplugged activities, craft prototypes |
| ~$20–30 | micro:bit board; Code & Go Robot Mouse |
| ~$60–100 | Bee-Bot, Sphero Indi/Mini, micro:bit + servo kits |
| $150+ | LEGO SPIKE / Mindstorms-class kits, VEX IQ, mBot |

**Tip for robotics teams:** your existing FRC/FTC robot is the best demo hardware you own — no purchase needed.

## Design & Engineering Tools

- **Tinkercad** — tinkercad.com. Free browser CAD; also has a circuits simulator. Class management for group accounts.
- **NASA STEM Engagement** — nasa.gov/stem. Free robotics and rover activities, plus Mars rover media that captivates every age group.
- **NASA Space Place** — spaceplace.nasa.gov. Younger-audience explainers on space robots.
- **Instructables** — instructables.com. Community build guides; screen projects for age-appropriateness first.
- **e-NABLE** — enablingthefuture.org. Volunteer network 3D-printing prosthetic hands — a powerful real-world example for the Cadette assistive-design challenge.

## Pathways & Programs to Recommend to Scouts

- **FIRST** — firstinspires.org. FIRST LEGO League (ages 9–14), FIRST Tech Challenge (12–18), FIRST Robotics Competition (14–18). Use the "team finder" to locate local teams.
- **Girls Who Code** — girlswhocode.com. Free clubs (grades 3–12) and free summer immersion programs.
- **Technovation Girls** — technovation.org. Global competition where girls build apps solving community problems.
- **EngineerGirl** — engineergirl.org. National Academy of Engineering site with role-model profiles, essay contests, and Q&A with real engineers.
- **NCWIT Aspirations in Computing** — aspirations.org. Award program for high-school women in computing (relevant for your mentors, too!).

## Research & Background for Leaders

- **Girl Scout Research Institute** — girlscouts.org/en/footer/research. Studies on girls in STEM — useful statistics for grant applications, presentations, and Gold Award "national/global link" write-ups.
- **NGCP (National Girls Collaborative Project)** — ngcproject.org. Statistics on the STEM gender gap and a directory of girl-serving STEM programs.
- **FIRST Impact Research** — firstinspires.org/about/impact. Longitudinal data showing outcomes for FIRST participants, especially girls.

## Running Your Own Workshop — Quick Checklist

1. **6+ weeks out:** contact troop leaders; confirm date, level(s), and headcount; leaders review VTK requirements
2. **4 weeks out:** recruit mentors (target 1:3–4 for Brownies, 1:4–5 for Juniors, 1:5–6 for Cadettes); reserve space with WiFi

3. **2 weeks out:** run a mentor training session using the mentor guides in each curriculum; test all laptops/tablets on the venue WiFi
4. **1 week out:** print materials (arrow cards, design sheets, checklists, surveys); confirm badge purchase with troop leaders
5. **Day of:** pre-surveys at check-in; follow the module schedules; post-surveys before pickup
6. **After:** tally survey results, send thank-yous, share the curriculum links, and log everything (hours, headcounts, badges earned)

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*Compiled for the "Girls Build Bots" Girl Scout Gold Award project. All external resources are free unless noted; verify links and age-appropriateness before your event. Official badge requirements © Girl Scouts of the USA, available in the Volunteer Toolkit.*